Class	Adm No

Candidate Name: ______________________________

H

2018 Preliminary Exams
Pre-university 3

BIOLOGY HIGHER 2 **9744/04**

Paper 4 Practical **20 September 2018**

Candidates answer on the Question Paper **2 hour 30 minutes**

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.
Give details of the practical shift and laboratory, where appropriate, in the boxes provided.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams and graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid/tape.

Answer all questions in the spaces provided on the Question Paper.

The use of scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

Shift
Laboratory

For Examiner's Use	
1	
2	
3	
Total	

This question paper consists of 15 printed pages, including 2 blank pages.

Answer **all** questions

1. The enzyme amylase catalyses the hydrolysis of starch to a reducing sugar.

You are required to identify which solution, **E1** or **E2** contains the highest enzyme concentration by estimating the concentration of reducing sugar produced by the action of **E1** and **E2** when breaking down starch.

The concentration of reducing sugar can be deduced by comparing with known concentration of reducing sugars. The reducing sugars are identified by using the Benedict's test.

You are provided with:

Labelled	**contents**	**hazard**	**Volume/cm^3**
E1	Amylase solution	Irritant Harmful	5
E2	Amylase solution	Irritant Harmful	5
S	Starch solution	None	50
G1	0.4% reducing sugar solution	None	50
W	Distilled water	none	100

Proceed as follows.

Read through step 1 to 7 and answer question a(i) to a(iii) before starting the experiment.

1 Put 1 cm^3 of **E1** and **E2** into separate beakers.

2 Put 6 cm^3 of **S** into each of the beakers with **E1** and **E2**. Mix well.

3 Leave the beakers aside for 15 minutes.
During the 15 minutes, prepare the known concentrations of reducing sugar solutions according to **a(i)**.

(a)

(i) Complete Fig. 1.1 to show how you will make three further concentrations of reducing sugar, **G2**, **G3** and **G4**. You should use serial dilution to reduce the concentration by half between each concentration.

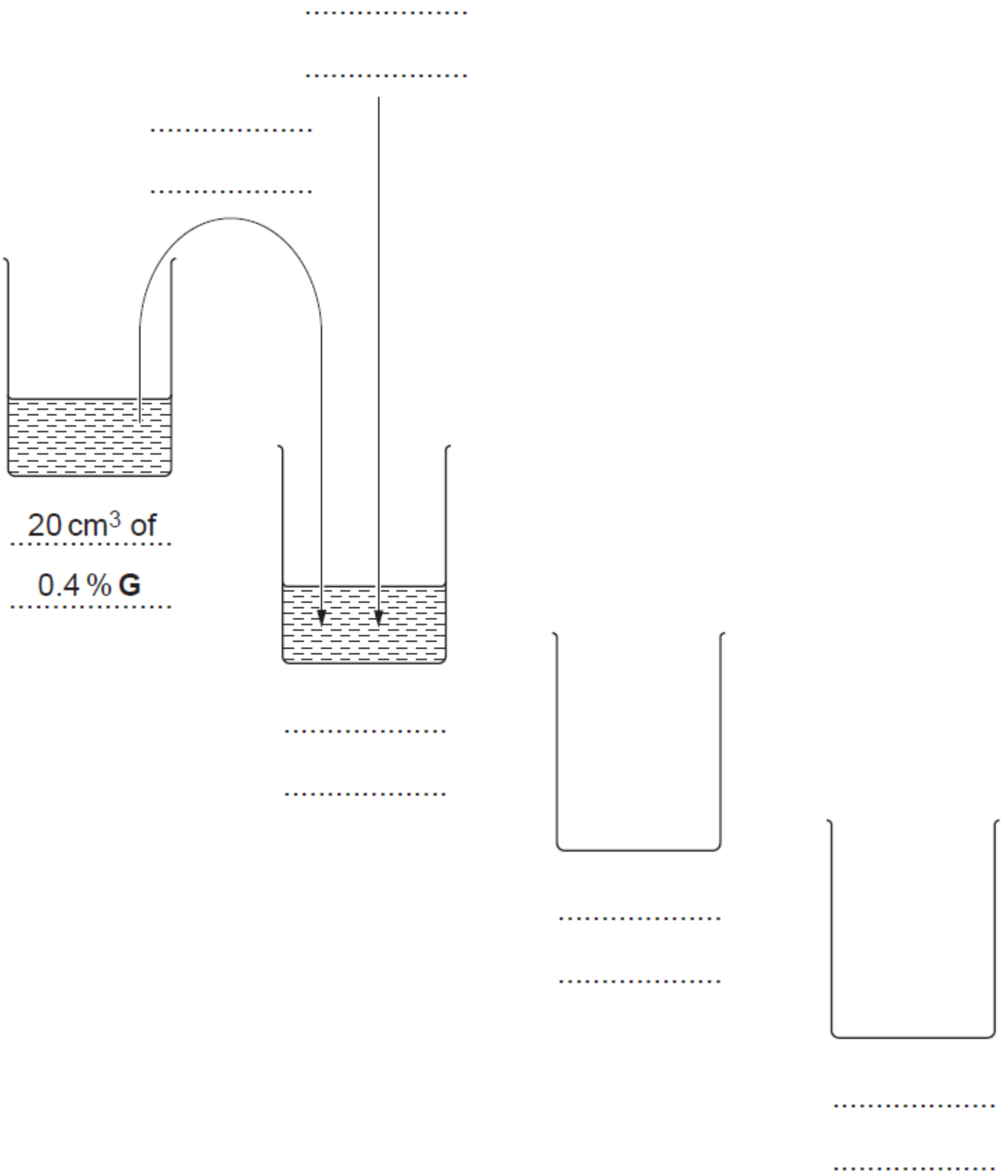

Fig. 1.1

[3]

- G2 0.2% + G3 0.1% + G4 0.05%;
- Transfers 10 from second beaker to third, third to fourth + cm^3;
- Adds 10 cm^3 distilled water/W to each of three beakers;

4 Prepare all the concentrations of the reducing sugar solution which are stated in Fig. 1.1, in the beakers provided.

5 You are now required to test for the concentrations of reducing sugar by using the Benedicts's test.

For each of your Benedict's test, you need to standardise the

- volume of Benedict's solution
- volume of the samples
- temperature of the water bath.

(ii) State the

- Volume of sample (max 6 cm^3) is same/lesser as volume of Benedict's solution + temperature 80 to 100 ^oC;

Volume of each of the samples to be tested …………………… cm^3

Volume of Benedict's solution …………………… cm^3

Temperature of water bath …………………… ^oC

[1]

6 Set up the water bath to heat to the temperature as decided in **a(ii).**

7 Test each sample with Benedict's test separately using the volumes decided. Record the time taken for the appearance of any colour change. If there is no colour change after 120 seconds, record 'more than 120'.

(iii) Prepare the space below and record the results for Benedict's test.

Samples	Time taken for colour change in Benedict's test(s)
G1	30
G2	50
G3	>120
G4	>120
E1	30
E2	70

- Correct heading for dependent variable: in first column, time in s and Correct heading for independent variable: percentage concentration of reducing sugar;
- Records readings for all glucose standards, E1 and E2 reactions, to the nearest seconds;
- Correct pattern of results for glucose standards where highest concentration has shorter time than next concentrations;
- Correct result for E1 (+/- 20 second):
- Correct result for E2 (+/- 20 second):

[4]

(iv) Using your results in **a(iii)**, state if **E1** or **E2** has higher concentration of reducing sugar.

- Correctly identified E1;

. **[1]**

(v) Using your results in **a(iii)**, estimate the concentration of reducing sugars in **E1** and **E2**.

- Correct % estimate for E1 and E2;

… **[1]**

(b) Plan an experiment to follow the time course of the hydrolysis of starch by amylase, without the use of Benedict's solution.

You may select form the following sterilized apparatus and plan to use appropriate additional apparatus:

- Normal laboratory glasswares, e.g. test-tubes, boiling tubes, beakers, measuring cylinders, graduated pipettes and pipette fillers, glass rods, etc.

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it.
- Be illustrated by relevant diagrams(s), if necessary
- Identify the independent and dependent variables
- Describe the method with the scientific reasoning used to decide the method so that the results are as accurate and repeatable as possible
- Use the correct technical and scientific terms

Max 3 for theory

1. Amylase is an enzyme that composes of active site that is complementary to the shape of starch, and binds to starch to form enzyme-substrate complex;
2. The rate of reaction can be compared by the amount of time required to digest starch into maltose which is indicated by the colour change of iodine from blue-black to yellow;
3. The shortest time indicate fastest rate;
4. Independent variable is the type of/concentration of amylase solution/E1 and E2 amylase solution and dependent variable is the time taken for iodine to give yellow result; [compulsory]
5. Controlled constant variable is pH, temperature, volume of reagents and concentration of reagent; (Min 3)
6. AVP;

Max 6 for procedure

1. Labelled diagram of set-up;
2. Conduct a pilot experiment to determine suitability of the apparatus, optimum conditions and amount of materials used;
3. Equilibrate starch and amylase solutions separately in 37°C water bath for 5 mins;
4. Add 5 cm^3 of starch and 2 cm^3 of amylase E1 into a test tube and mix well. Incubate for 15 minutes;
5. Use a pipette and add ten drops of iodine across a white tile;
6. Label each spot with a marker as 1st 30s, 2nd 30s,…. 10th 30s;
7. After the 15 minutes incubation, take drop of mixture and drop it on the white tile every 30 second interval, for a total of 300 seconds/5 minutes;
8. Stop the experiment when the iodine solution started turning from brown/orange to blue-black/remain yellow;
9. Repeat step 4 to 8 with amylase E2;
10. A control is set up with boiled amylase to ensure that the reaction is due to the activity of amylase;
11. To ensure reliability of results, repeat steps 2 to 9 to obtain a total of three readings/triplicates, and calculate the average;
12. To ensure reproducibility of data, repeat the entire experiment twice using freshly prepared reagents and solutions;

(c) In a separate experiment, the effect of iron sulfate on the rate of amylase activity was investigated. Amylase was immobilised in alginate beads. Two types of alginate beads containing amylase was prepared – with or without iron sulfate.

The two types of beads were mixed together in varying proportions. Fig. 1.2 shows an example of 30 beads with iron sulfate mixed with 70 beads without iron sulfate (30% with iron sulfate). The beads were placed into beakers containing starch solution.

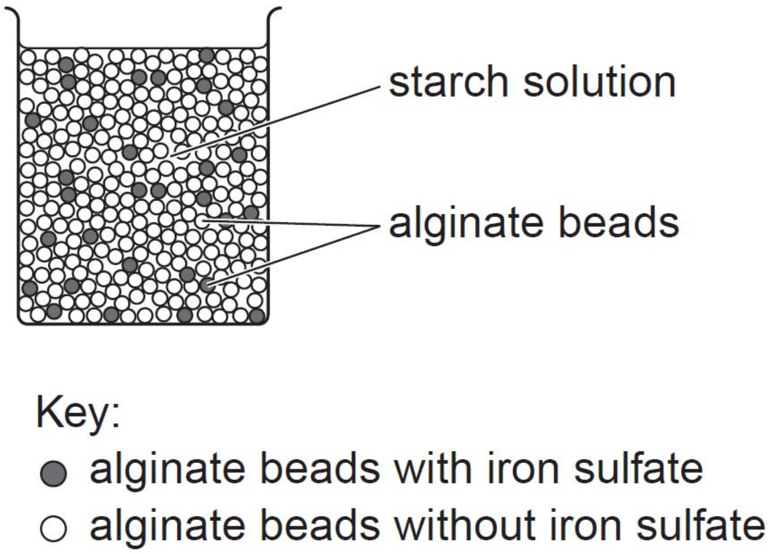

Fig. 1.2

The mass of reducing sugar produced by each mixture of beads in one minute was measured. The results are shown in Table 1.1.

Table 1.1

Percentage of beads with iron sulfate/%	Mass of reducing sugar produced/ µg
0	60
10	25
20	22
30	5
40	2
50	0

(i) Plot the data in Table 1.1 in the grid below.

- Axes correctly labelled with correct units;
- Scaled appropriately with ascending scale and equidistant intervals, with a scale such that plotted points occupy at least 50% of the graph paper in both the x and y directions;
- All values correctly plotted;
- Smooth best fit line/graph;

[4]

(ii) State the anomalous data and explain your answer.

- The anomalous data is at 20% of beads with iron sulfate;
- As it does not fit within the best fit line/trend/AVP;

[2]

(iii) Describe two constant variables in this experiment.

- Temperature of the experiment to ensure that reaction is not limited by low/insufficient kinetic energy/denaturation/AVP;
- Size of alginate beads to ensure constant amount/mass of iron sulfate/AVP;
- Volume of starch solution to ensure that all set-ups have the same volume of substrate/AVP;
- Concentration of starch solution to ensure that all set-ups have the same concentration of substrate/AVP;
- Concentration of amylase solution to ensure all set-ups have the same concentration of enzyme;

(iv) Explain the reason for the difference in the results between 0% and 10% of beads with iron sulfate.

……………………………………………………………………………………………

- Iron sulfate inhibits the rate of enzyme-catalysed reaction;
- alters active site of enzyme to produce lesser enzyme-substrate complexes/ lesser amount of product per unit time;
- fewer starch can bind to the active site of enzyme, producing lesser amount of glucose;

……… **[2]**

[Total: 29]

[Turn over

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2. You are required to investigate the light-dependent stage in photosynthesis in a leaf extract using the dye DCPIP. When DCPIP is reduced, it changes from blue colour to colourless.

You are provided with:

Labelled	contents	hazard	Volume/cm^3
L1	Fresh leaf	None	-
L2	Cold buffer solution	None	15
L3	Sand	None	20
L4	0.05% DCPIP	None	15

Proceed as follows.

1 Place the piece of fresh leaf, **L1**, on a white tile.

2 Using a scissor, carefully remove any large veins and discard them. Cut the rest of the leaf into small pieces and place them into a plastic container.

3 Using a pipette, add 3 cm^3 of cold buffer solution, labelled **L2** into the plastic container.

4 Add a spatula of sand, labelled **L3** into the plastic container. Carefully grind the chopped leaf using a glass rod for about one minute to obtain a green leaf extract.

5 Clean the surface of the white tile and place a Petri dish on the edge of the tile as shown in Fig. 2.1. Ensure that the Petri dish is tilted.

6 Pour the leaf extract from the plastic container into the Petri dish and allow it to drain to the edge of the dish as shown in Fig. 2.1.

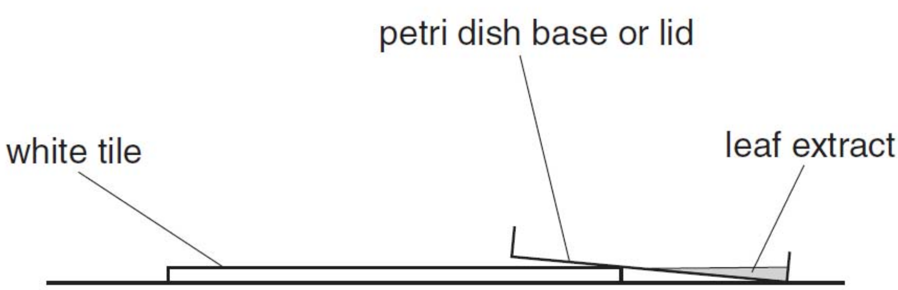


Fig. 2.1

7 Completely cover the Petri dish with aluminum foil to shield the content from light. Ensure that the foil is made easy to cover and remove.

You must cover the Petri dish with the foil at all times, except when adding reagent or removing samples.

8 Insert one end of a capillary tube into the leaf extract to collect some leaf extract. Remove the capillary tube and lay it on the white tile. **Label this tube as Tube 1**.

9 Using a pasteur pipette, add four drops of 0.05% DCPIP solution to the leaf extract in the Petri dish. Ensure the two liquids are evenly mixed to form a blue solution. If not, add more drops of 0.05% DCPIP to achieve a blue solution.

10 Take a second capillary tube and collect some leaf extract/DCPIP solution. Wrap this tube with aluminum foil to prevent any exposure to light. Place the capillary tube on the same white tile. **Label this tube as Tube 2.**

11 Take a third capillary tube and collect some leaf extract/DCPIP solution. Place the capillary tube on a separate white tile. Place the white tile and this capillary tube near a bench lamp. **Label this tube as Tube 3.**

12 Carefully check the colour of the liquid in all tubes every minute, for the next ten minutes. Note the time taken for the **DCPIP solution** to change from blue to colourless. If there is no change in colour after 10 minutes, stop the investigation.

(a) Draw a table to show your results.

tube	time taken for colour change (minutes)
1	➢ 10
2	➢ 10
3	4

- Correct heading for dependent variable: in first column, tube and Correct heading and units for independent variable: time taken for DCPIP to change from blue to colourless in minutes;
- Records readings for all three tubes;
- Record Tube 1 as > 10 mins, tube 2 taking more time than tube 3;

[3]

(b) Suggest the role of **Tube 1.**

- Control to ensure that the results in tube 2 and tube 3 is due to the presence of DCPIP;

.....[1]

(c) Using your own knowledge, explain the expected results in relation to the light-dependent stage of photosynthesis.

- In the light dependent reaction of photosynthesis, the photon of light excites the chlorophyll;
- Boost the electrons to a higher energy level and is accepted by electron acceptor;
- DCPIP acts as electron acceptor to absorb/accept electrons and is reduced;
- Instead of NADP that is being reduced to reduced NADP;
- Tube 3 which is exposed to light will take shorter time to change colour;

.....[4]

(d) Suggest the purpose of using the buffer solution, **L2** in step 3.

- Maintain constant pH;
- Maintain suitable osmotic equilibrium/environment;
- AVP;

.....[1]

(e) Describe two risks involved in this experiment and suggest how you could take precautions against them.

- Scissor may cause cut when slicing the leaf, so handle with care to avoid cut/AVP;
- Glass rod may break when grinding the leaf, so do not apply too much force/AVP;
- Glassware/Petri dish/capillary tube is fragile, so handle with care to avoid breakage and cut;

.....[2]

(f) Suggest three ways the investigation could be improved to make the results more reliable.

- Filter/centrifuge the leaf extract to obtain the supernatant;
- Maintain constant temperature for all tubes throughout experiment by using thermostatically controlled environment/bath;
- Use colourimeter to measure light intensity;
- Repeat experiment;
- AVP;

......**[3]**

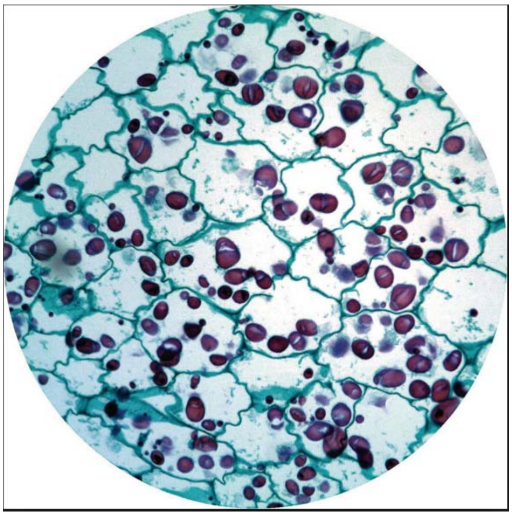
[Total: 14]

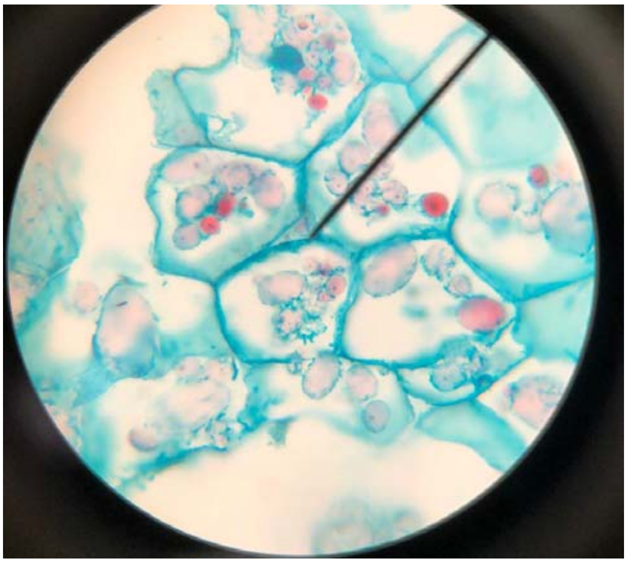
Novel

3. Slide **S1** is a slide showing the transverse section of a plant stem. This plant is an important crop plant throughout the world.

(a) Observe and select a group of four cells. Each cell in the group should touch two other cells.

Make a large detailed drawing of this group of four cells. Use one ruled label line to identify the component of the cell.





- Clear continuous line with no overlap and cellulose cell wall shown;
- Drawing of 4 cells, in proportionate size;
- Label and annotate with straight horizontal line (cell wall, cell surface membrane);
- Label and identify starch granules in cell;

[4]

(b)

(i) Use the eyepiece graticule and the stage micrometer to measure the actual diameter of each of the four cells that you have drawn in **(a)**.

Identify each cell by writing the letters **A** to **D** on your drawing in **(a)**.

Show your working in the space below.

- Calibration of stage micrometer with eyepiece;
- Indicate cell A, B, C and D in drawing with straight horizontal labelled line;
- Correct diameter of cell A in correct unit;
- Correct diameter of cell B in correct unit;
- Correct diameter of cell C in correct unit;
- Correct diameter of cell D in correct unit;

diameter of cell ***A*** ....

diameter of cell ***B*** ....

diameter of cell ***C*** ....

diameter of cell ***D*** ....

[5]

(ii) Calculate the mean diameter of cell **A** to **D** and the standard deviation. Use the formula for standard deviation.

$$s = \sqrt{\frac{\Sigma(x - \overline{x})^2}{n - 1}}$$

Show your working in the space below.

- Correct calculation of mean diameter in correct unit;
- Correct working using formula;
- Correctly identify n = 4;
- Correct final answer;

[3]

[Total: 12]